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Flank pain in the third trimester as a clue to diagnose spontaneous adrenal hemorrhage

Busara Songtanin, MD, Nicole Welch, MD, Kenneth Nugent, MD, and Anupa Patel, MD

Department of Internal Medicine, Texas Tech University Health Sciences Center, Lubbock, Texas

ABSTRACT

Flank pain in pregnancy is a serious complaint and may have several etiologies, such as pyelonephritis or trauma. Spontaneous adrenal hemorrhage is rare in pregnancy, but is life threatening. We report a third-trimester pregnant woman with idiopathic bilateral spontaneous adrenal hemorrhage who presented with acute-onset severe flank pain and developed adrenal insufficiency. Normal blood pressure following hemorrhagic infarction may obscure the underlying adrenal insufficiency.

KEYWORDS Adrenal hemorrhage; adrenal insufficiency; flank pain; pregnancy

Bilateral adrenal hemorrhage in pregnancy is exceedingly rare. The incidence and underlying pathophysiology in pregnancy is not well understood. Here, we present a case of adrenal insufficiency following spontaneous bilateral adrenal hemorrhage in the third trimester in a patient who presented with flank pain and nausea.

CASE DESCRIPTION

A 27-year-old primigravida woman at gestational age 32 weeks presented to an outside hospital with sudden-onset severe left flank and back pain associated with nausea for 4 hours. The pain radiated to the left anterior abdomen and was stabbing in nature and 8/10 in severity. She denied fever, dysuria, diarrhea, and recent trauma. She had positive fetal movements and no rupture of membranes. She denied a previous history of abdominal pain and had normal antenatal care. Vital signs showed a temperature of 36.6°C, heart rate of 94 beats/min, respiratory rate of 22 breaths/minute, and blood pressure of 108/49 mm Hg. Physical examination revealed tenderness and left lower quadrant pain along with left flank pain.

Laboratory tests showed a hemoglobin of 12.3 g/dL and white cell count of 14.4 k/μL. An electrolyte panel showed sodium, 127 mmol/L; potassium, 3.8 mmol/L; bicarbonate, 11 mmol/L; and chloride, 108 mmol/L. Blood cultures were negative. She had a normal coagulation profile and negative lower-extremity Dopplers. Urinalysis was normal. Ultrasound of the abdomen was negative for ureteric stones and hydronephrosis.

The patient was transferred to our care due to concerns of adrenal hemorrhage. Upon arrival, she complained of severe bilateral flank pain. Computed tomography (CT) of the abdomen and pelvis showed enlargement (3.1 × 2.8 cm) and slight hyperattenuation of the left adrenal gland with loss of its adreniform shape. There was thickening of the right adrenal gland possibly due to mild hemorrhage *(Figure 1)*. Laboratory tests reported a random adrenocorticotropic hormone (ACTH) level of 11.3 pg/mL (reference range 7.2–63.3) and a random cortisol of 2.8 mcg/dL (low). The patient was given intravenous fluids and hydrocortisone for 6 days and then switched to prednisolone 20 mg daily before doing an ACTH stimulation test. Cosyntropin stimulation tests were low at baseline (0.8 mcg/dL; reference range 5.0–21.0), at 30 minutes (1.0 mcg/dL; reference range 14.0–36.0), and at 60 minutes (1.3 mcg/dL; reference 14.0–41.0). The diagnosis of adrenal insufficiency was made, and the patient was started on hydrocortisone 20 mg twice daily and fludrocortisone 0.1 mg daily until delivery. At follow-up postpartum, she was mildly hypertensive, and the steroid dose was tapered to hydrocortisone 30 mg daily and fludrocortisone 0.05 mg daily.

Corresponding author: Busara Songtanin, MD, Department of Internal Medicine, Texas Tech University Health Sciences Center, 3601 4th Street, Lubbock, TX 79430 (email: busara.songtanin@ttuhsc.edu)

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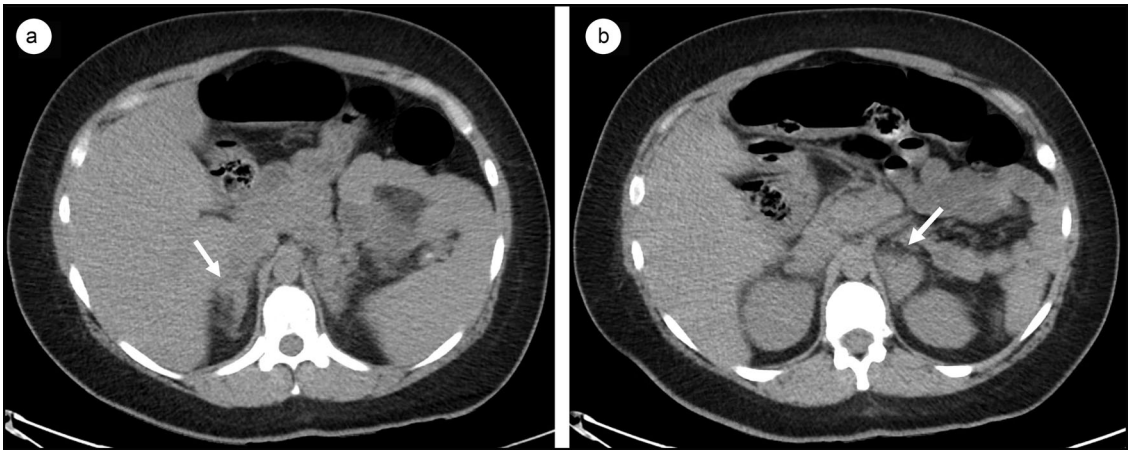


Figure 1. CT showing **(a)** loss of adreniform shape with slight hyperattenuation in the right adrenal gland, and **(b)** loss of adreniform shape in the left adrenal gland, with the gland measuring 3.1 × 2.8 cm likely due to hemorrhage.

DISCUSSION

Spontaneous adrenal hemorrhage is a very rare condition that requires prompt diagnosis and treatment. It often presents with nonspecific signs and symptoms, which makes a high index of suspicion critical for the rapid diagnosis of this potentially fatal condition. The most common causes of adrenal hemorrhage are infection and sepsis. However, trauma, anticoagulant use, autoimmune diseases, and pregnancy are also known causes.[1] The exact pathophysiology is unknown, but speculation has produced several theories.[2] The adrenal glands have a rich vascular supply. Since pregnancy is a stress on the body, the secretion of ACTH can lead to adrenal hypertrophy with an increased vascular supply, thus increasing the risk of venous congestion, which may ultimately lead to hemorrhage.[1,2]

The presentation of spontaneous adrenal hemorrhage often includes nonspecific symptoms. In cases of adrenal hemorrhage with adrenal insufficiency, patients can present with fatigue, dizziness, weakness, myalgias, fever, nausea, vomiting, and diarrhea.[3] On initial evaluation, patients may be hypertensive due to the rapid release of cortisol and catecholamines during hematoma formation but will quickly become tachycardic and/or hypotensive due to the decreased levels of the hormones.[2]

CT is the most important tool for the diagnosis of spontaneous adrenal hemorrhage. Magnetic resistance imaging is more accurate and can determine if a tumor is present or if the hemorrhage is chronic.[2,4] Initial management should include fluid resuscitation, vasopressor support if necessary, corticosteroid therapy, and correction of any underlying cause. If patients fail conservative therapy and have persistent bleeding, surgery or embolization may be indicated.[3] Even with prompt recognition and treatment, the mortality rate is 15%.[5]

ORCID

Busara Songtanin http://orcid.org/0000-0002-4117-4272
Kenneth Nugent http://orcid.org/0000-0003-2781-4816

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